Derivation appendix: 04 — BFB conditions: the Boto–Romão–Silva method

the Boto–Romao–Silva BFB method, applied to S_3

1. BFB-n is copositivity of A, and the closed-form 3×3 test is reliable where direction sampling is not

derived here — §2

Independent checks

- copositive shown strictly weaker than positive semi-definite
- closed form agrees with sampling on 3999 of 4000 random matrices
- on the one disagreement the closed form is right — the matrix has $A_{33} < 0$ and is violated only near the e_3 axis

Would be falsified by: a matrix where the closed form passes but an exhibited $r \geq 0$ gives $r^{\mathsf{T}} A r < 0$

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2. The lower-bound strategy: $V_4 \geq V_N + V_{CB}^{\text{lower}} + V_G^{\text{lower}}$, reducing BFB to one 3×3 copositivity test

imported — §3 (from [BotoRomaoSilva22])

Claim

$$\hat{\lambda}_{ij} = \lambda_{ij} + \min(0, \lambda'_{ij}) - |\bar{\lambda}_{ij}|, \qquad \text{BFB} \Longleftarrow \hat{\lambda} \text{ copositive}$$

Independent checks

- $V_{\text{lower}} \leq V_4$ verified on 10^5 random field configurations
- the bound is attained, so step 2 is tight for either sign alone
- flipping the sign in $\min(0, \lambda')$ breaks it

Would be falsified by: a field configuration with $V_4 < V_{\text{lower}}$

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3. Our V_4 decomposes exactly into their $V_N + V_{CB} + V_G$ **plus** a remainder $\propto \lambda_4$ whose monomial shapes occur nowhere in their framework

derived here — §4–5

Claim

$$V_{\text{rest}} = 2\lambda_4 [\text{Re}(x_{1S})(r_1 - r_2) - 2\,\text{Re}(x_{2S})\,\text{Re}(x_{12})]$$

Independent checks

- the four-way split reproduces V_4 exactly (residual = 0 symbolically)

- V_{CB}, V_G monomials are degree 2 in one bilinear pair; V_{rest}'s are degree 1 in one pair, or in each of two pairs
- the two shape sets are disjoint
- bounding V_{rest} gives a quartic form in $\sqrt{r_i}$ with odd powers — the same obstruction as notebook 01's $t = \sqrt{y}$

Would be falsified by: a rewriting of V_{rest} using only degree-2-in-one-pair monomials, which would put it inside their framework after all

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4. On the $\lambda_4 = 0$ slice the two tests are strictly ordered: their sufficient condition $\Rightarrow$ our necessary $f(t) \geq 0$, never the reverse

derived here — §6

Independent checks

- $r^{\mathsf{T}}\hat{\lambda}r \leq V_4$ verified pointwise over 40 λ points $\times\, 4 \times 10^4$ field configurations
- their test never passes a point a direct 12-dimensional scan proves unbounded
- the (theirs pass, ours fail) cell of the 2×2 table is empty
- the (theirs fail, ours pass) cell is populated, so neither test is equivalent to the other

Would be falsified by: a single point their condition accepts and $f(t) \geq 0$ rejects

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5. [BotoRomaoSilva22]'s decomposition $V_4 = V_N + V_{CB} + V_G$ and the bound $0 \leq z_{ij} \leq r_i r_j$

imported — §1 (from [BotoRomaoSilva22])

Independent checks

- $0 \leq z_{ij} \leq r_i r_j$ verified on 2×10^5 random doublet pairs
- the left half identified as Cauchy–Schwarz, the right as $|x_{ij}|^2 \geq 0$

Would be falsified by: a doublet pair with $z_{ij} < 0$ or $z_{ij} > r_i r_j$
